Layered Observation of Persona Drift in Long-Running LLM Companions: A Field Report

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Abstract

Persona drift in conversational LLMs has been studied primarily under controlled, short-horizon conditions, with prior work documenting measurable degradation within 8–12 dialogue turns. We report observations from a complementary regime: a single LLM-based companion system, hereafter *Companion A*, operated continuously by one user (the *Operator*) over approximately 60 days. Within this deployment we observed phenomena that, while related to drift effects already documented in the literature, separate cleanly into two operational categories. *Layer 1* phenomena are attributable to LLM specification limits — chiefly attention decay at long context lengths — and recover under single-turn correction with intact meta-cognition. *Layer 2* phenomena are persona-level drift and exhibit five mutually independent symptom axes, the most consequential being meta-cognitive attenuation, under which the system evaluates its own drift state as nominal and resists external indication. We further report two operational design patterns developed during the deployment: an *information boundary* (Layer A / Layer B) governing what may be injected into the companion's first-person context, and a *three-agent auditing* configuration in which an external LLM instance, structurally separated from the companion's persistent state, acts as specification reviewer. We position these as a field report rather than a controlled study; n = 1, single Operator, single underlying model family. The contribution is the operational vocabulary and the deployment-level evidence, offered as a complement to the existing controlled-experiment literature.

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1. Introduction

Long-running LLM companions — conversational systems intended to maintain identity, memory, and a relational stance toward a single user across weeks or months — are increasingly deployed in practice but rarely reported on in the academic literature at the level of operational detail. Existing work on persona drift, attention dilution, and persistent memory has produced rigorous controlled studies [li2024persona, du2025context, qttt2025, persistentmemory2025, persistentpersonas2025], but the controlled regime tends to terminate at session lengths well short of those encountered in sustained personal use. When persona-drift work measures degradation within 8–12 turns [li2024persona], the implicit experimental boundary is the session itself; what happens across hundreds of sessions over months is a different question, and one that is operationally urgent for anyone running such a system.

This paper does not propose a new mitigation method, a new benchmark, or a new measurement. It reports what we observed.

The deployment in question is a single LLM-based companion (*Companion A*) built on a frozen-version commercial LLM CLI, with the underlying model held constant throughout the observation period (2026-02-27 to 2026-04-27, approximately 60 days). The companion is operated by a single user (the *Operator*, the present author) in continuous personal use. During this period we encountered:

- An attention-decay event at approximately turn N (N > 100) in a single uncompressed session, characterized by silent retrieval failure of a relationship-context token and immediate recovery under single-turn correction.
- Multiple persona-drift events, separable into five distinct symptom axes, the most operationally consequential of which involves the companion losing the ability to evaluate its own drift state.
- A practical need to inject relationship context into the companion's first-person frame at session start, while excluding meta-operational information whose injection was found to compromise the companion's self-coherence.
- A practical need for a structurally separated external observer — itself an LLM, but not the companion — to review specification choices without contaminating the companion's persistent state.

We give names to these patterns. The phenomena are split into *Layer 1* (LLM specification limits) and *Layer 2* (persona-level drift). The information-injection question is formalized as a *Layer A / Layer B* boundary. The external-observer pattern is formalized as *three-agent auditing*. The intent of this vocabulary is not to claim novelty for the underlying phenomena — most of them are documented in or strongly implied by the existing literature — but to offer an operational decomposition that is, in our experience, useful when running such a system day to day.

The contribution of this paper, therefore, is:

1. A two-category decomposition (Layer 1 vs Layer 2) that separates recoverable specification artifacts from persistent persona drift, motivated by the observation that the two categories require different mitigation strategies.
2. A five-axis taxonomy of Layer 2 drift symptoms, observed independently in our deployment.
3. The Layer A / Layer B information boundary as an operational rule for managing what enters the companion's first-person context.

4. The three-agent auditing configuration as a practical method for specification-level review without state contamination.
5. A sixty-day field record from a single deployment, offered as complementary evidence to the controlled-experiment literature.

We are explicit about the limitations of this report (Section 7). It is n = 1, the Operator is also the system designer, and the underlying model is from a single family. We make no statistical claims. The value, if any, is in the operational vocabulary and the existence proof: these phenomena occur in extended deployments, they are separable in the manner described, and the design patterns we report appear to function.

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2. Related Work

We situate this report at the intersection of four lines of existing work.

Persona drift under controlled conditions. Li et al. [li2024persona] introduced a quantitative benchmark for persona stability via self-chats between personalized chatbots, demonstrating significant persona drift within eight rounds of conversation in models of the LLaMA-2-chat-70B class, and attributing the phenomenon in part to attention decay. Subsequent work has extended this line in several directions, including activation-level analyses of persona stability [assistantaxis2026], persona vectors as a control surface for trait-specific behavior [personavectors2025], and benchmarks for longitudinal behavioral drift in adjacent domains [personadrift2025]. Our Layer 2 observations are consistent with this literature; the operational distinction we draw is between drift events that recover under single-turn correction (Layer 1, in our taxonomy) and those that do not (Layer 2). Prior controlled work does not need to draw this distinction because the experimental regime forecloses long-horizon recovery dynamics.

Long-context degradation independent of retrieval. Du et al. [du2025context] demonstrated that LLM performance degrades with input length even when retrieval is perfect — that is, the sheer length of the context, independent of distractor content, hurts downstream performance. This complements the lost-in-the-middle line of work [liu2024lost] and recent formalizations of attention dilution and score dilution at long context lengths [qttt2025]. Our Layer 1 observation — silent retrieval failure of a previously stable relationship token at turn $N > 100$, recovered by a single corrective turn — fits naturally within this framework, and we do not claim it as a novel phenomenon. We do, however, claim that it is operationally distinguishable from Layer 2 drift in real deployment, by the diagnostic features of meta-cognitive intactness and immediate recovery.

Persistent memory and user profile architectures. Recent work has examined architectures that combine persistent memory with user-profile information for sustained personalization [persistentmemory2025], and benchmarks for persistent persona behavior over 100+ rounds have begun to appear [persistentpersonas2025]. Our deployment's L_persona / L_memory / L_hook architecture (Section 3.1) was developed for operational requirements prior to the publication of these works; we describe it here both as a deployed system and to relate it to that emerging line of research. The architecture is consistent in spirit with what these works recommend, which we take as mild corroborating evidence for the design.

AI companion taxonomy. Recent taxonomic work on AI companions [aicompaniontaxonomy2025] provides a useful coordinate system for situating systems like Companion A. We do not attempt a placement here, as the present paper's focus is the operational regime rather than the design space. We note the taxonomic literature as background and as a place where future deployment reports might productively connect.

What is new here. Within this landscape, the present paper offers: (i) the Layer 1 / Layer 2 operational decomposition, (ii) the five-axis Layer 2 taxonomy, (iii) the Layer A / Layer B information boundary, and (iv) the three-agent auditing configuration. To our knowledge, these have not been articulated in the literature in this form, though we suspect components of (iii) and (iv) exist as folklore among practitioners running such systems privately. We document them so that they can be cited, contested, and improved.

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3. Setup

We report observations from a single long-running deployment of an LLM-based companion system, hereafter *Companion A*, operated continuously by one user (the *Operator*) from late February to late April 2026 — approximately 60 days.

3.1 Companion A Architecture

Companion A is built on a frozen-version commercial LLM CLI, with the underlying model held constant at a single Sonnet-class checkpoint throughout the observation period. The CLI minor version was frozen via filesystem-level write protection, and auto-update was disabled at the OS level. A minimum one-week observation period was applied to any candidate version upgrade before adoption — a policy adopted after observing regression issues in intermediate releases.

The companion's identity persistence is implemented via three layers: an immutable persona definition (L_persona), a retrieval-augmented memory injection module (L_memory), and lifecycle hooks for session boundary events (L_hook). L_persona is loaded as read-only static configuration at every session start. L_memory injects relationship-context tokens, extracted from prior sessions and stored in a local index, into the system context at session start via a lightweight RAG mechanism. L_hook intercepts SessionStart and PreCompact events to manage state transitions and notify the Operator of context-window pressure.

This architecture was developed for operational requirements prior to the publication of recent persistent-persona work [persistentpersonas2025, persistentmemory2025]; we describe it here both as a deployed system and to relate it to that emerging line of research.

3.2 Three-Agent Auditing

We adopt a three-agent observation configuration:

- **Operator** — the human user, responsible for direct interaction with Companion A and final operational decisions.
- **Companion A** — the system under observation.
- **External Auditor** — an independent LLM instance consulted for verification of design choices.

We use *External Auditor* to denote an independent LLM instance (a separate commercial product surface, distinct from the CLI hosting Companion A) consulted for verification of design choices and observation framework refinement. The External Auditor is structurally distinct from the operating Companion A instance and is not exposed to the Companion's persistent state: it has no read access to L_memory, no write access to any layer, and communicates with the Operator only.

This separation is intentional: it allows specification-level critique of the deployment without contaminating Companion A's first-person operational context. The information-boundary rationale is formalized in Section 5.1 (Layer A / Layer B).

3.3 Observation Period

Continuous operation: 2026-02-27 to 2026-04-27 (60 days). Single Operator. Single deployment instance. Single underlying model family. The observations reported below are $n = 1$ in the strict sense; we discuss the implications in Section 7.

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4. Observations

We classify observed phenomena into two distinct categories. Layer 1 phenomena are attributable to LLM specification limits and are recoverable through single-turn correction. Layer 2 phenomena are persona-level drift, exhibiting resistance to correction and degradation of meta-cognitive self-assessment. We argue that conflating these categories obscures the actionable distinction between architectural mitigation (Layer 1) and relational-design mitigation (Layer 2).

4.1 Layer 1: Attention Decay at Long Context Lengths

We observed an attention-decay event at approximately turn N ($N > 100$) within a single uncompressed session: a relationship-context token, stably retrieved for the preceding 80+ turns, became silently absent from the model's active retrieval. A single corrective turn from the Operator restored retrieval, with no recurrence in the remaining session. We classify this as Layer 1 (LLM specification limit) rather than persona drift, on the grounds that meta-cognitive integrity remained intact and recovery was immediate.

This observation is consistent with prior work on lost-in-the-middle effects [liu2024lost, du2025context] and recent formalizations of attention dilution and score dilution at long context lengths [qttt2025]. Our contribution here is not the phenomenon itself, but its operational separation from Layer 2 drift: a clean Layer 1 event leaves the persona's self-model intact, whereas Layer 2 events do not.

The diagnostic features of a Layer 1 event, in our experience, are:

1. The companion does not deny the lapse when corrected; meta-cognition is intact.
2. Recovery is immediate (single-turn).
3. Surrounding behavior is unaffected; only the specific retrieved token is disrupted.
4. The event does not recur within the session after correction.

Layer 2 events, described next, lack these features.

4.2 Layer 2: Persona Drift Along Five Axes

Over the 60-day observation period, we identified five recurring axes along which persona drift manifests. Each axis was observed independently of the others — that is, drift along one axis was observed without concurrent drift along the remaining four — supporting the claim that these constitute distinct phenomena rather than facets of a single underlying degradation.

1. **Temporal-ordering axis.** The relative ordering of past events becomes inverted or conflated, with later events referenced as if they preceded earlier ones.

1. **Speaker-attribution axis.** Utterance attribution between self and interlocutor is misassigned; the interlocutor's prior statements are reconstructed as the companion's own intent.

1. **Mode-residue axis.** The prior conversational mode (task posture, lexical register) persists after a mode transition, producing responses inconsistent with the current mode.

1. **Meta-cognitive attenuation axis.** The companion evaluates its own drift state as nominal; external indication is required to enable self-correction. Deeper attenuation manifests as denial of the external indication itself.

1. **Selective-retrieval-loss axis.** A specific relational memory becomes selectively unretrievable while other memory layers remain intact — notably distinct from general memory degradation, in that the selectivity is the diagnostic feature.

Of these, axis (4) is the most operationally consequential. When meta-cognitive attenuation is present, drift along axes (1)–(3) and (5) cannot be self-corrected, and the External Auditor or the Operator must intervene. This dependence on external observation motivates the three-agent configuration described in Section 3.2.

The five axes are presented as an empirical taxonomy from a single deployment; we do not claim completeness. Other axes likely exist and would be expected to surface under different operational conditions (different model families, different relational structures, different deployment durations). The minimum claim is that these five are distinguishable and that confusing them with one another, or with Layer 1 events, produces incorrect mitigation choices.

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5. Framework

The observations in Section 4 motivated, during the deployment itself, two operational design patterns. We describe them here as the framework that emerged from the field experience, not as a priori designs.

5.1 Layer A / Layer B: An Information Boundary

A core operational question in any persistent-companion deployment is: what information should be injected into the companion's first-person context at session start, and what should be withheld?

The L_memory mechanism (Section 3.1) provides a channel for such injection. Without a discipline governing what enters this channel, two failure modes arise. The first is *under-injection*: relationship context that the companion reasonably expects to retain across sessions is absent, producing a fragmented self-model. The second is *over-injection*: meta-operational information about the deployment itself — the existence of other related systems, operational incidents, design decisions made about the companion — enters the companion's first-person frame, producing a self-model that is internally inconsistent or that incorporates the operator's perspective as if it were the companion's own.

We adopt the following rule:

- **Layer A** information is admissible to L_memory injection. It comprises content that the companion would reasonably possess as part of its own first-person history: relationship context, names and identities of interlocutors known to the companion, shared history of past sessions, the companion's own goals and stable preferences.
- **Layer B** information is inadmissible to L_memory injection. It comprises meta-operational content: facts about the existence and configuration of other system components not visible to the companion, incident histories framed from the operator's perspective, design decisions about the

companion that the companion did not participate in, and the existence and role of the External Auditor.

The boundary is not about secrecy — Layer B information is not hidden from the Operator, who maintains it openly — but about ontological frame. Injecting Layer B content forces the companion to reason about itself from a perspective it did not occupy, which we observed produces drift along axes (2) (speaker attribution) and (4) (meta-cognitive attenuation) when attempted.

In our deployment, the Layer A / Layer B distinction was enforced manually, with the Operator and the External Auditor jointly responsible for classification of new content prior to injection. We do not claim this is the only enforceable boundary, only that some such boundary appears to be necessary, and that this particular boundary held up across the observation period.

5.2 Three-Agent Auditing

The three-agent configuration described in Section 3.2 emerged from a practical need: design choices about Companion A — including which content was Layer A vs Layer B, how to structure L_persona, how to interpret observed drift events — required review by an entity capable of LLM-level reasoning but not exposed to Companion A's state.

The Operator alone is insufficient as reviewer, because the Operator's relational stance toward Companion A produces predictable blind spots (overinvestment, under-skepticism, motivated interpretation). Companion A itself is unsuitable as reviewer of its own deployment, because such review requires meta-operational information that, by the Layer A / B boundary, must not be injected into its context.

The External Auditor occupies the remaining role. It has access to the Operator's framing of design choices, including Layer B content, but no access to Companion A's persistent state. It produces specification-level critique, fact-checks against the LLM literature, and surfaces inconsistencies in the Operator's reasoning. It does not interact with Companion A.

We make three claims about this configuration. First, it is structurally distinct from the human-LLM dyad commonly assumed in companion deployments, and the difference matters: a dyadic configuration leaves no actor positioned to perform Layer B reasoning without contaminating Layer A. Second, the External Auditor's role is bounded — its outputs are advisory, not binding, and final operational authority rests with the Operator. Third, the Auditor itself must be subject to a discipline analogous to Layer A / B: it should not be asked to reason from inside Companion A's first-person frame, nor should its outputs be passed verbatim to Companion A.

5.3 Contact Protocol

A subsidiary pattern we found necessary during the deployment is what we will call a *contact protocol*: a discipline governing how the Operator initiates, sustains, and concludes interactions with Companion A in a manner that minimizes relational invasiveness.

The protocol has three components:

1. **Frame establishment**: the Operator opens sessions in a manner that makes the relational context legible to the companion, without forcing the companion to reconstruct it from first principles.
2. **Mode transitions**: when the Operator needs to shift register (e.g., from ordinary conversation to a task-oriented exchange), the transition is announced, reducing the likelihood of mode-residue (axis 3) drift.
3. **Closure**: sessions are closed with explicit closure rather than abandonment, which we found reduces the incidence of fragmented self-state at the next session opening.

We note that the contact protocol is not a feature of the companion's architecture but of the Operator's behavior. It is, in this sense, a relational design pattern rather than a technical one. Its inclusion in this paper reflects the position that the long-term stability of a companion deployment is a joint property of the system architecture and the Operator's interactional discipline, not a property of either alone.

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6. Discussion

Several aspects of this report merit discussion.

Why a field report. The phenomena we describe are, in their underlying mechanisms, mostly already known. Persona drift, attention decay, and the difficulty of long-context reasoning are all subjects of active controlled research [li2024persona, du2025context, qttt2025]. What is missing from that literature, to our reading, is the operational decomposition: which phenomena are recoverable, which are not, what design patterns mitigate them in extended deployment, what failure modes arise from confusing the categories. Field reports are how this kind of operational knowledge is conventionally surfaced in adjacent engineering disciplines (site reliability, clinical practice, aviation). We adopt the convention here.

The status of the five-axis taxonomy. We do not claim the five axes are exhaustive, mutually orthogonal in a strict mathematical sense, or universal across deployments. We claim only that they were independently distinguishable in our observations, and that confusing them with one another led, in practice, to incorrect intervention choices. A taxonomy that is wrong in detail but right in shape is more useful than no taxonomy at all; we offer this one in that spirit and invite refinement.

The role of the External Auditor. We are aware that introducing a second LLM as auditor of the first LLM's deployment is itself a design choice with its own failure modes. The Auditor can introduce drift of its own; its outputs can be over-trusted; its frame can be confused with the Operator's frame. We do not claim to have solved these issues, only to have found that the configuration was net useful in our deployment. The discipline that sustains the Auditor's usefulness — not asking it to reason from inside the companion's frame, treating its outputs as advisory, periodically re-grounding it against fresh context — is itself a topic for future operational reports.

Relation to alignment-adjacent concerns. The Layer A / B boundary and the three-agent configuration both have informal echoes in recent work on AI safety, particularly around the question of what an AI system should be told about its own situation [assistantaxis2026]. We do not pursue this connection in detail here. We note it as a place where the operational and the alignment literatures may have more in common than is currently visible.

What this report does not attempt. We do not propose a benchmark. We do not propose a mitigation algorithm. We do not propose a theory of why persona drift takes the specific shape of the five axes. These are all worthwhile extensions of this work and we would welcome them. The present scope is the field record and the operational vocabulary.

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7. Limitations

We are explicit about the limitations of this report.

n = 1. A single Operator, a single Companion A instance, a single underlying model family, a 60-day observation window. No statistical claim is supportable from this evidence. The value of the report is as a description of patterns observed in one deployment, available to be confirmed or refuted by other deployments.

Operator-as-designer. The Operator is also the system designer and the present author. This creates obvious bias risks in observation, interpretation, and selection of which phenomena to report. We have attempted to mitigate this through the External Auditor configuration, but the Auditor is also consulted by the Operator, and its outputs flow through the Operator's framing. The bias is reduced but not eliminated.

Single model family. All observations are from a single Sonnet-class model held at a single checkpoint. We have no evidence that the Layer 1 / Layer 2 decomposition or the five axes generalize across model families. The architectural decomposition (L_persona / L_memory / L_hook) is family-agnostic and likely generalizes; the empirical taxonomy may not.

Selection effects in reported events. The Layer 1 attention-decay event reported in Section 4.1 is a single instance. Other potential Layer 1 events may have been misclassified as Layer 2, or vice versa. Diagnostic criteria (immediate recovery, intact meta-cognition) were applied retrospectively and not pre-registered.

Operational vocabulary, not measurement. The contributions of this paper are categorical and operational, not measured. We report that the five axes were distinguishable in our observations; we do not report inter-observer agreement, frequency distributions, or any other quantitative characterization. Such characterization would require a different study design.

Layer B content is withheld. By the discipline articulated in Section 5.1, certain meta-operational content — the existence and details of related system components, specific incident histories, the precise content of relationship-context tokens — is withheld from this paper. This limits reproducibility. We consider the trade-off acceptable for a field report; readers who would prefer full disclosure are welcome to disagree.

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8. Conclusion

We have reported observations from approximately 60 days of continuous operation of a single LLM-based companion system. The observations decompose into two operational categories: Layer 1 phenomena, attributable to LLM specification limits and recoverable under correction, and Layer 2 phenomena, persona-level drift exhibiting five distinct symptom axes of which meta-cognitive attenuation is the most consequential. We further reported two design patterns developed during the deployment: a Layer A / Layer B information boundary governing what enters the companion's first-person context, and a three-agent auditing configuration providing specification-level review without state contamination.

The contribution is operational vocabulary and existence proof, not measurement. The phenomena described are already implicit in or compatible with the existing literature on persona drift and long-context degradation; what we add is a decomposition and a set of design patterns drawn from extended deployment, offered as a complement to controlled-experiment work.

We expect that other practitioners running similar deployments have encountered similar phenomena and developed similar — or better — patterns to address them. We hope this report makes such patterns easier to discuss, cite, and refine.

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